for each row i,

Compute the <u>scale</u> of each row

$$s_i = \max_{1 \le j \le n} |a_{ij}| = \max\{|a_{i1}|, |a_{i2}|, \dots, |a_{in}|\}$$

Select pivot row for which

$\frac{|a_{i1}|}{s_i}$ is the largest

The row index chosen is denoted by P_1 → switch equations to make P_1 the 1^{st} row.

Next, carry out a round of elimination to obtain 0s in the first column of A.

Then continue computing scales of the rows $P_2, P_3 \dots P_n$ and rearranging the equations as above.

Ex.

$$\left[\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 12 \\ 12 & -8 & 6 & 10 & 34 \\ 3 & -13 & 9 & 3 & 27 \\ -6 & 4 & 1 & -18 & 38 \end{array}\right.$$

Scales: $S_1 = 6$ $\quad \frac{|a_{11}|}{S_1} = 1$

$S_2 = 12$ $\quad \frac{|a_{21}|}{S_2} = 1$

$S_3 = 13$ $\quad \frac{|a_{31}|}{S_3} = \frac{3}{13}$

$S_4 = 18$ $\quad \frac{|a_{41}|}{S_4} = \frac{6}{18}$

$P_1 = 1.$

Reduction

$$\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 12 \\ 0 & -4 & 2 & 2 & 10 \\ 0 & -12 & 8 & 1 & 21 \\ 0 & 2 & 3 & -14 & -26 \end{array}$$

Scales: $\frac{|a_{22}|}{S_2} = 1$, $\frac{|a_{32}|}{S_3} = 1$

$$\frac{|a_{33}|}{S_3} = \frac{2}{14}$$

$P_2 = 2$

Reduction

$$\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 12 \\ 0 & -4 & 2 & 2 & 10 \\ 0 & 0 & 2 & -5 & -9 \\ 0 & 0 & 4 & -13 & -21 \end{array}$$

Scales:

$$\frac{|a_{33}|}{s_3} = \frac{2}{5} \qquad \frac{|a_{43}|}{s_4} = \frac{4}{13}$$

$$\frac{2}{5} > \frac{4}{14} \Rightarrow p_3 = 3$$

Reduction

$$\left[\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 12 \\ 0 & -4 & 2 & 2 & 10 \\ 0 & 0 & 2 & -5 & -9 \\ 0 & 0 & 0 & -3 & -3 \end{array}\right]$$

Back substitution yields

$$\bar{x} = \begin{bmatrix} 1 \\ -3 \\ -2 \\ 1 \end{bmatrix}.$$

Operation Counts

- Focus only on multiplications (divisions) since they are significantly more time-consuming than additions/subtractions.

Consider the computational cost estimation for scaled row pivoting G. E.

<u>Factorization phase</u>

- to define p_1 (the index of the pivot row), need n divisions

$$\boxed{\frac{|a_{i1}|}{s_i} \quad i = 1, 2, \dots, n}$$

- for rows $p_2, \dots, p_n$, compute a multiplier (1 op), then subtract a multiple of row p_1 from rows $p_2, \dots, p_n$ $\longrightarrow$ n ops per row

→ $n(n-1)$ ops total

total so far $\simeq n(n-1)+n = n^2$ ops.

- Remainder of the process works the same way, except the matrices ~~become~~ progressively smaller.

Total ops for the fact. phase

$$= n^2 + (n-1)^2 + (n-2)^2 + \dots + 2^2 =$$

$$= \frac{1}{6} n(n+1)(2n+1) - 1 \simeq \frac{1}{3} n^3 + O(n^2)$$

as $n \to \infty$

<u>Solution phase</u>

- Transform b using the same row operations as used in the factorization phase →

→ $n-1$ ops on 1^{st} pass

→ $n-2$ on the 2^{nd} et.

...

1 ...

Total for updating $\bar{b}$:

$$= (n-1) + (n-2) + \dots + 1 = \frac{1}{2} n(n-1)$$

In the back-substitution part, solving for x_n takes 1 op, solving for x_{n-1} takes 2 ops, and so on

The total here is

$$1 + 2 + 3 + \dots + n = \frac{1}{2} n(n+1)$$

Thus the total for the solution phase is

$$\frac{1}{2} n(n-1) + \frac{1}{2} n(n+1) = n^2 + O(n)$$

as $n \to \infty$.

(!!) This (sensible) implementation of G.E. takes about $\frac{1}{3} n^3$ ops as $n \to \infty$.

Diagonally-dominant matrices.

D.d. condition

$$\boxed{|a_{ii}| > \sum_{\substack{j=1\\ j\neq i}}^{n} |a_{ij}| \quad (i = 1, 2, \ldots, n)}$$

Ex. $A = I = \begin{pmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & 1 & 0 & \ldots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \end{pmatrix}$

$|a_{ii}| = 1$, $1 > 0 \to$ A is diagonally dominant.

Thm Gaussian elimination without pivoting preserves diagonal dominance.

Pf. Enough to prove this only for step 1 (other steps are similar).

We have to show:

$$\left|a_{ii}^{(2)}\right| > \sum_{\substack{j=2 \\ j\neq i}}^{n} \left|a_{ij}^{(2)}\right| \qquad i = 2,3,\dots n$$

$$\Updownarrow$$

$$(*)\quad \left|a_{ii}^{(2)}\right| = \left|a_{ii} - \frac{a_{i1}}{a_{11}} a_{1i}\right| \overset{?}{>} \sum_{\substack{j=2 \\ j\neq i}}^{n} \left|a_{ij} - \frac{a_{i1}}{a_{11}} a_{1j}\right|$$

We will prove a stronger inequality

$$(**)\quad |a_{ii}| - \left|\frac{a_{i1}}{a_{11}} a_{1i}\right| \overset{?}{>} \sum_{\substack{j=2 \\ j\neq i}}^{n} |a_{ij}| + \left|\frac{a_{i1}}{a_{11}} a_{1j}\right|$$

$\forall\, a,b \in \mathbb{R}$: $\quad |a+b| \le |a| + |b|$ - triangle inequality

$\big||a| - |b|\big| \le |a-b|$ - "reverse" Δ inequality

By reverse triangle,

$$|a-b| \geq |a| - |b|$$

$$\left|a_{ii} - \frac{a_{i1}}{a_{11}} a_{1i}\right| \geq |a_{ii}| - \left|\frac{a_{i1}}{a_{11}} a_{1i}\right|$$

By triangle,

$$\sum \left| a_{ij} - \frac{a_{i1}}{a_{11}} a_{1j} \right| \leq \sum |a_{ij}| + \sum \left| -\frac{a_{i1}}{a_{11}} a_{1j} \right|$$

$$= \sum \left(|a_{ij}| + \left| \frac{a_{i1}}{a_{11}} a_{1j} \right| \right)$$

(**) has the smaller LHS and larger RHS than (*). Thus (**) $\Rightarrow$ (*) (but conversely).

From diag. dominance,

$$|a_{ii}| - \sum_{\substack{j=2 \\ j\neq i}}^{n} |a_{ij}| > |a_{i1}| \qquad i=2,\dots,n$$

So, to get $(**)$, it is enough to prove that

$$(***) \quad |a_{i1}| \overset{?}{\geq} \sum_{j=2}^{n} \left|\frac{a_{i1}}{a_{11}} a_{1j}\right| = |a_{i1}| \underbrace{\sum_{j=2}^{n} \left|\frac{a_{1j}}{a_{11}}\right|}_{<1}$$

Diag. dominance in row 1:

$$|a_{11}| > \sum_{j=2}^{n} |a_{1j}| \iff$$

$$1 > \sum_{j=2}^{n} \frac{|a_{1j}|}{|a_{11}|}$$

$$\Downarrow$$

$(***)$ is true $\Rightarrow$ $(**)$ holds $\Rightarrow$ $(*)$ holds. $\square$

<u>Corollary 1</u>. if $A_{n \times n}$ is d.-d., then

A is non-singular and has an

LU-factorization.

<u>Corollary 2</u> no pivoting is required

for G.E. of d.-d. matrices

Norms and Error Analysis. 4.4

$\bar{x}$ - an element of a linear vector space V

Def norm: is a function

$$\bar{x} \to \|\bar{x}\|$$

$$\|\bar{x}\| \in \underbrace{\mathbb{R}^{+} \cup \{0\}}_{\text{non-negative real \#s.}}$$

Satisfying:

1. $\|\bar{x}\| > 0$ if $\bar{x} \neq 0$, $\forall \bar{x} \in V$.
2. $\|\lambda \bar{x}\| = |\lambda| \|\bar{x}\|$, $\forall \lambda \in \mathbb{R}$, $\forall \bar{x} \in V$
3. $\|\bar{x}+\bar{y}\| \leq \|\bar{x}\| + \|\bar{y}\|$, $\forall \bar{x}, \bar{y} \in V$

Note: 2. implies that $\|0\| = 0$.

Indeed if $\bar{x} = 0$, and $\lambda \neq 0$ then $\|\lambda \cdot 0\| = |\lambda| \|0\|$

$$\Downarrow$$

$$\|0\| = |\lambda| \|0\| \Rightarrow \|0\| = 0.$$

also

$$\|-\bar{x}\| = \|(-1)\bar{x}\| = |-1|\,\|\bar{x}\| = \|\bar{x}\|$$

$\forall \bar{x} \in V,\quad \bar{x} \neq 0.$

$\boxed{\text{Ex}}$ Euclidean norm on $\mathbb{R}^n$

Def $\|\bar{x}\|_2 =$

$$= \left(\sum_{i=1}^{n} x_i^2\right)^{1/2}$$

$(V = \mathbb{R}^n$

$\bar{x} \in \mathbb{R}^n \Leftrightarrow$

$\bar{x} = (x_1, x_2, \dots, x_n)^T$

Other norms on $\mathbb{R}^n$:

$$\|\bar{x}\|_\infty = \max_{i=\overline{1,n}} |x_i|$$

$$\|\bar{x}\|_1 = \sum_{i=1}^{n} |x_i|$$

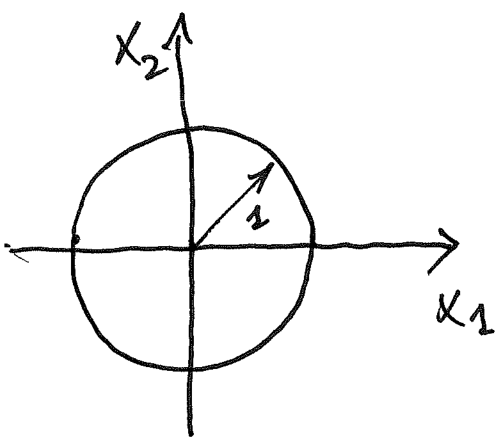
$$\|\bar{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}} \qquad p \geq 1$$

Ex) $V = \mathbb{R}^2$

Graph the sets $\|\bar{x}\|_2 \leq 1$,

$\|\bar{x}\|_\infty \leq 1$,

$\|\bar{x}\|_1 \leq 1$.



1) $\|\bar{x}\|_2 \leq 1 \iff$

$(x_1^2 + x_2^2)^{1/2} \leq 1 \quad \forall x_1, x_2 \in \mathbb{R}$

$\downarrow$

disk of radius 1 centered at zero

2) $\|\bar{x}\|_\infty \leq 1 \iff \max_{i=1,2} |x_i| \leq 1 \iff$

$\iff \max\{|x_1|, |x_2|\} \leq 1$

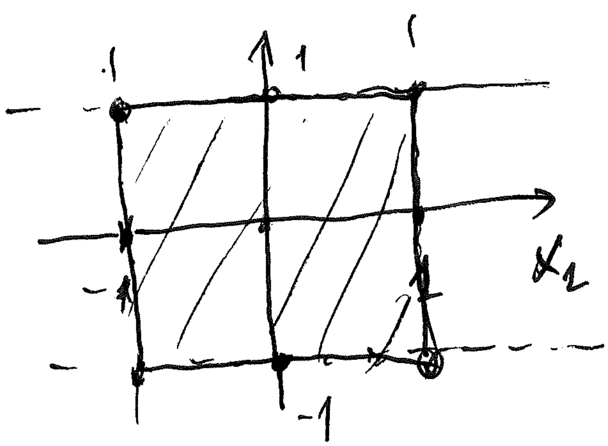
$|x_1| \leq 1$

$\Updownarrow$

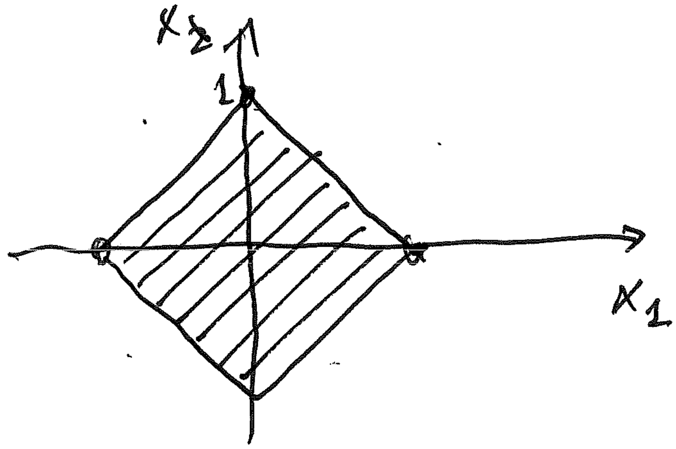
$-1 \leq x_1 \leq 1$

$|x_2| \leq 1$

$-1 \leq x_2 \leq 1$



3) $\|\bar{x}\|_1 \leq 1 \iff |x_1| + |x_2| \leq 1$



boundary:

$|x_1| + |x_2| = 1.$

quad. I

$x_1 \geq 0, \quad x_2 \geq 0$

$x_1 + x_2 = 1$

$x_2 = -x_1 + 1$

quad. II $\quad x_1 \leq 0, \quad x_2 \geq 0$

$$|x_1| + |x_2| = -x_1 + x_2 = 1$$

$$x_2 = x_1 + 1$$

other quadrants – similar.

<u>Note:</u> If a norm $\|\cdot\|$ is defined on V, then $\|\bar{x} - \bar{y}\|$ for any $\bar{x}, \bar{y} \in V$ may be used as a measure of the distance between $\bar{x}$ and $\bar{y}$.

Thm If $\|\cdot\|$ is any norm on $\mathbb{R}^n$, then the equation

$$\|A\| = \sup_{\|\bar{u}\|=1} \{ \|A\bar{u}\| : \bar{u} \in \mathbb{R}^n \} \quad (*)$$

defines a norm in the linear space of all $n\times n$ matrices.

Pf. 1) If $A \neq 0$, than A has ~~at~~ a non-zero column $A^{(j)} \neq 0$

let $\bar{x} = (0, \ldots 0, \overset{(j)}{1}, 0 \ldots 0)^T$,

$$\bar{v} = \frac{\bar{x}}{\|\bar{x}\|} \to \|\bar{v}\| = \left\| \frac{\bar{x}}{\|\bar{x}\|} \right\| = \frac{1}{\|\bar{x}\|} \|\bar{x}\| = 1$$

Now $\|A\| \geq \|A\bar{v}\| =$

$$= \left\| A \frac{\bar{x}}{\|\bar{x}\|} \right\| = \left\| \frac{1}{\|\bar{x}\|} A\bar{x} \right\| =$$

$$= \frac{1}{\|\bar{x}\|} \|A\bar{x}\| = \frac{\|A^{(j)}\|}{\|\bar{x}\|} > 0$$

2) $\|\lambda A\| = \sup_{\|\bar{u}\|=1} \{ \|\lambda A \bar{u}\| : \|\bar{u}\|=1 \} =$

$= \sup_{\|\bar{u}\|=1} \{ |\lambda| \, \|A\bar{u}\| ; \|\bar{u}\| = 1 \} =$

$= |\lambda| \sup_{\|\bar{u}\|=1} \{ \|A\bar{u}\| \} = |\lambda| \, \|A\|$

done.

3) $\|A+B\| \leq \sup_{\bar{u}} \{ \|(A+B)\bar{u}\| ; \|\bar{u}\|=1 \} \leq$

Δ on $\|\cdot\|$

$\leq \sup \{ \|A\bar{u}\| + \|B\bar{u}\| : \|\bar{u}\|=1 \} \leq$

$\leq \sup \{ \|A\bar{u}\| : \|\bar{u}\|=1 \} + \sup \{ \|B\bar{u}\| ; \|\bar{u}\|=1 \}$

$= \|A\| + \|B\|$

$\square$

Note: $\|A\bar{x}\| \leq \underbrace{\|A\|}\cdot\|\bar{x}\|$ (**)

Indeed: if $\bar{x}=0$, then $\|A\bar{x}\|=0$

$\|A\|\|\bar{x}\|=0$ - ok.

if $\bar{x}\neq 0$, consider $\bar{v}=\dfrac{\bar{x}}{\|\bar{x}\|}$, $(\|\bar{v}\|=1)$

$$\|A\| \geq \|A\bar{v}\| = \frac{\|A\bar{x}\|}{\|\bar{x}\|}$$

$$\Downarrow$$

$$\|A\|\|\bar{x}\| \geq \|A\bar{x}\|$$

Other (special) properties of a subordinate matrix norm.

a) $\|I\|=1$

b) $\|AB\| \leq \|A\|\|B\|$ (***)

To show (***), consider $\bar{u}\in\mathbb{R}$, $\|\bar{u}\|=1$.

$$\|AB\bar{u}\| \overset{(**)}{\leq} \|A\|\|B\bar{u}\| \overset{(**)}{\leq} \|A\|\|B\|\|\bar{u}\| = \|A\|\|B\|$$

Now, for any $\bar{u}$: $\|\bar{u}\| = 1$,

$$\|AB\bar{u}\| \le \|A\|\|B\|$$

$$\Downarrow$$

$$\sup_{\bar{u}:\|\bar{u}\|=1} \|AB\bar{u}\| \le \|A\|\|B\|$$

$$\Updownarrow$$

$$\|AB\| \le \|A\|\|B\|.$$

$\square$.

Ex. Matrix norm subordinate to $\|\bar{x}\|_\infty$ on $\mathbb{R}^n$.

$$\|A\|_\infty = \sup_{\|\bar{u}\|_\infty=1} \|A\bar{u}\|_\infty = \sup_{\|\bar{u}\|_\infty=1} \{\max_{i=1,2,\dots,n} |(Au)_i|\} =$$

$$= \max_{i=1,2,\dots,n} \sup_{\|\bar{u}\|_\infty=1} |(Au)_i| = \max_{i} \sup_{\|\bar{u}\|_\infty=1} \left|\sum_{j=1}^{n} a_{ij} u_j\right| \le$$

$$\le \max_{i} \sup_{\|\bar{u}\|_\infty \le 1} \sum_{j=1}^{n} |a_{ij}| \overbrace{|u_j|}^{\le 1}$$

$$\leq \max_{i} \underset{\|\bar{u}\|=1}{\overset{\text{drop.}}{\sup}} \sum_{j=1}^{n} |a_{ij}| = \underbrace{\max_{i=1,\dots,n} \sum_{j=1}^{n} |a_{ij}|}_{\text{in terms of } a_{ij}}$$

Double
Check that

$$\sup_{\|\bar{u}\|_\infty = 1} \left| \sum_{j=1}^{n} a_{ij} u_j \right| = \sum_{j=1}^{n} |a_{ij}|$$

$$\|\bar{u}\|_\infty = 1 \iff \max_{j=\overline{1,n}} |u_j| = 1 \Rightarrow$$

$$-1 \leq u_j \leq 1 \qquad j = 1, 2, \dots, n$$

if $a_{ij} > 0$, put $u_j = 1$

if $a_{ij} < 0$ put $u_j = -1$

Then $|a_{ij} u_j| = a_{ij} u_j = |a_{ij}| \Rightarrow$

$$\left| \sum_{j=1}^{n} a_{ij} u_j \right| = \left| \sum_{j=1}^{n} |a_{ij}| \right| = \sum_{j=1}^{n} |a_{ij}|.$$

Ex, # 4 from 3.4

a) show that $f(x) = (1+x^2)^{-1}$ is contractive on an arb. interval

Solution: pick $I \subset \mathbb{R}$.

need: $|f(x) - f(y)| \le k\,|x-y| \quad \forall x, y \in I.$

$0 < k < 1$

Note that $f'(x) = -(1+x^2)^{-2} \cdot 2x =$

$= -\dfrac{2x}{(1+x^2)^2}$ exists $\forall x \in I$

MVT: (Taylor): $f(y) = f(x) + f'(\xi)(y-x)$

for some ξ between x and y

$\Downarrow$

$f(y) - f(x) = f'(\xi)(y-x)$

$|f(y) - f(x)| = |f'(\xi)|\,|y-x|$

$$|f(x) - f(y)| \leq \max_{\xi \in I} |f'(\xi)| \, |y - x|$$

if $0 < \ldots < 1$ then we have contractivity.

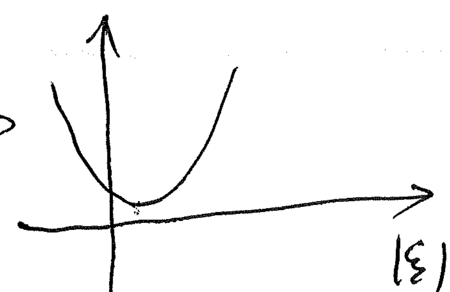
Consider $f'(\xi) = -\dfrac{2\xi}{(1+\xi^2)^2}$

$$|f'(\xi)| = \frac{2|\xi|}{(1+|\xi|^2)^2} = \frac{2|\xi|}{1+|\xi|^2} \cdot \frac{1}{1+|\xi|^2}$$

$$\frac{2|\xi|}{1+|\xi|^2} \overset{?}{\leq} C \quad \text{for some } C \in \mathbb{R}^+$$

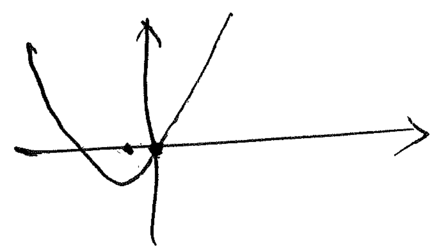
$$2|\xi| \overset{?}{\leq} C(1 + |\xi|^2)$$

$$C|\xi|^2 - 2|\xi| + C \overset{?}{\geq} 0$$



Need: real solutions of $C|\xi|^2 - 2|\xi| + C = 0$ must be both ≤ 0 (if they exist)

Or at least



quad. formula

$$|\xi| = \frac{2 \pm \sqrt{2^2 - 4c^2}}{2c} = \frac{2 \pm \sqrt{4-4c^2}}{2c} = \frac{2 \pm 2\sqrt{1-c^2}}{2c} \Leftrightarrow$$

interested in c small, $\Rightarrow$ can assume

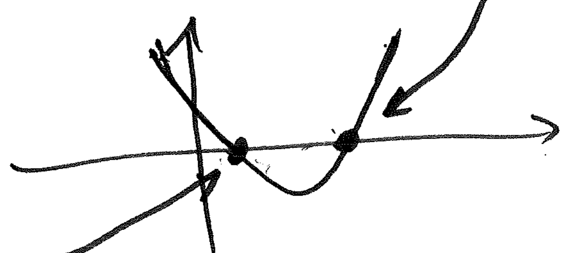
$1-c^2 > 0 \Leftrightarrow c^2 < 1 \Leftrightarrow -1 < c < 1$

Suppose $c > 0$

~~Both 2 ± 2~~

$\Leftrightarrow \frac{1 \pm \sqrt{1-c^2}}{c} \Rightarrow$ solutions

$\frac{1-\sqrt{1-c^2}}{c}$, $\frac{1+\sqrt{1-c^2}}{c}$ - both solutions > 0



Other option - have $c > 1 \rightarrow$ no real solutions, but as small as possible.

Ex $f(x) = \dfrac{1}{1+x^2}$

Show that $f(x)$ is contractive on any interval $I \subset \mathbb{R}$.

$$f'(x) = -\frac{2x}{(1+x^2)^2}$$

$$|f(x) - f(y)| \le \cancel{\max}\ \sup_{\xi \in I} \left| -\frac{2\xi}{(1+\xi^2)^2} \right| |x-y| =$$

~~for some~~

$$= \sup_{\xi \in I} \frac{2|\xi|}{(1+|\xi|^2)^2} |x-y|$$

Consider $g(\xi) = \dfrac{2\xi}{(1+\xi^2)^2}$, $\xi \geq 0$,

$$g(\xi) = \underbrace{\frac{2\xi}{1+\xi^2}}_{\text{not small enough for all } \xi} \cdot \frac{1}{1+\xi^2}$$

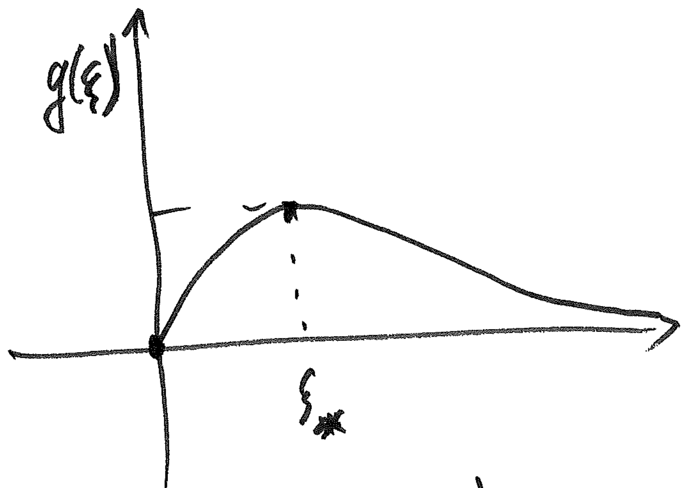
~~Ste~~ What is the behavior of the graph of $g(\xi)$?

$\lim_{\xi \to \infty} g(\xi) = 0$ $\quad$ 0 is a hor. asymptote

$g(0) = 0$

$g(\xi) > 0$ for $\xi > 0$

Conjecture:



Can we maximize $g(\xi)$ on $[0, \infty)$ using calculus?

$$g'(\xi) = \left[2\xi(1+\xi^2)^{-2}\right]' = 2\left\{(1+\xi^2)^{-2} + \xi \cdot (-2)(1+\xi^2)^{-3} \cdot 2\xi\right\}$$

$$= 2(1+\xi^2)^{-3}\left[1+\xi^2 - 4\xi^2\right] = 2(1+\xi^2)^{-3}(1-3\xi^2)$$

C. point(s): solutions of

$$1-3\xi^2=0$$

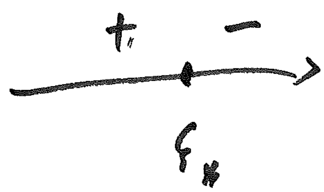
$$\xi^2=\frac{1}{3}$$

$$\Downarrow$$

$$\xi=\pm\frac{1}{\sqrt{3}}$$

only $\xi_* = \frac{1}{\sqrt{3}}$ matters.

Sign change of $g'(\xi)$ near ξ_* -?



$\Rightarrow$ local max at ξ_* (the only extremum)

abs. maximum on $[0,\infty)$ is

$$g(\xi_*) = g\left(\frac{1}{\sqrt{3}}\right) = \frac{2\cdot\frac{1}{\sqrt{3}}}{(1+3)^2} = \frac{2}{16\sqrt{3}} = \frac{1}{4\sqrt{3}}$$

$$g(\xi_*) < 1.$$

Thus,

$$|f(x) - f(y)| \le \underbrace{\frac{1}{4\sqrt{3}}}_{<1} |x-y|$$

$\Rightarrow$ $f(x)$ is a contraction.

$\boxed{\text{Ex}}$. 4.1 # 12

$A_{n\times n}$ - invertible, $u, v \in \mathbb{R}^n$.

Consider

$$\underset{(n+1)\times(n+1)}{\tilde{A}} = \begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

Find conditions on u, v that ensure invertibility of $\tilde{A}$ and find the inverse formula (when it exists).

1. Find conditions for invertibility

Use: $\tilde{A}\, x = 0 \Rightarrow x = 0$

Write $x = \begin{bmatrix} \bar{x} \\ x_n \end{bmatrix}$ $\bar{x} = (x_1, \dots x_n)^T$

Consider

$$\tilde{A} \begin{bmatrix} \bar{x} \\ x_n \end{bmatrix} = \begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix} \begin{bmatrix} \bar{x} \\ x_n \end{bmatrix} = \begin{bmatrix} A\bar{x} + x_n u \\ \cdots\cdots \\ v^T \bar{x} \end{bmatrix} = 0$$

$$A\bar{x} + x_n u = 0 \qquad (1)$$

$$v^T \bar{x} = 0 \quad \text{(scalar)} \qquad (2)$$

$$A\bar{x} = -x_n u$$

$$A^{-1} A \bar{x} = A^{-1}(-x_n u)$$

$$\Downarrow$$

$$\bar{x} = -x_n A^{-1} u$$

Plug into (2):

$$v^T(-x_n A^{-1} u) = 0$$

$$-x_n \underbrace{(v^T A^{-1} u)} = 0$$

$$\boxed{\text{Condition: } s := v^T A^{-1} u \neq 0} \qquad (*)$$

$(*) \Rightarrow \quad x_n = 0 \quad \Rightarrow$ Plug into (1)

$$A\bar{x} = 0 \Rightarrow \bar{x} = 0 \quad (A \text{ is invertible})$$

$$\Downarrow$$

$\tilde{A}$ is invertible iff $s \neq 0$.

2. Find formula for the inverse.

Seek

$$\hat{A}^{-1} = \begin{bmatrix} B & \hat{u} \\ \hat{v}^T & k \end{bmatrix}$$

$B_{\cancel{(n-1)} n \times n}$

$\hat{u}, \hat{v} \in \mathbb{R}^n$

$k \in \mathbb{R}$

Need to find $B, \hat{u}, \hat{v}, k$

given A, u, v from

$$\begin{array}{cc} A_{11} & A_{12} \\ \multicolumn{2}{c}{\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}} \\ A_{21} & A_{22} \end{array} \begin{array}{cc} B_{11} & B_{12} \\ \multicolumn{2}{c}{\begin{bmatrix} B & \hat{u} \\ \hat{v}^T & k \end{bmatrix}} \\ B_{21} & B_{22} \end{array} = \begin{bmatrix} I_n & 0 \\ 0 & 1 \end{bmatrix}$$

$n \times n$

Using the thm on products of block matrices, we have

$$A_{11}B_{11} + A_{12}B_{21} = I_n,$$
$$A_{11}B_{12} + A_{12}B_{22} = 0,$$
$$A_{21}B_{11} + A_{22}B_{21} = 0,$$
$$A_{21}B_{12} + A_{22}B_{22} = 1.$$

$$\Downarrow$$

$$AB + u\hat{v}^T = I_n, \qquad (3)$$
$$A\hat{u} + ku = 0, \qquad (4)$$
$$v^TB + 0\cdot\hat{v}^T = 0, \qquad (5)$$
$$v^T\hat{u} + 0\cdot k = 1. \qquad (6)$$

$(4) \Rightarrow \quad \hat{u} = -kA^{-1}u$. Plug into (6)

$$v^T(-kA^{-1}u) = 1 \Leftrightarrow -k(v^TA^{-1}u) = 1$$

$$\Leftrightarrow -ks = 1 \Rightarrow$$

$$\boxed{k = -\frac{1}{s}} \quad (7)$$

Now

$$\boxed{\hat{u} = -k A^{-1} u = \frac{1}{s} A^{-1} u} \quad (8)$$

Next, from (3),

$$AB = I_n - u \hat{v}^T$$

$$B = A^{-1}(I_n - u \hat{v}^T)$$

Plug into (5):

$$v^T A^{-1} (I_n - u \hat{v}^T) = 0$$

$$v^T A^{-1} - (v^T A^{-1} u) \hat{v}^T = 0$$

$$v^T A^{-1} - s \hat{v}^T = 0$$

$$\hat{v}^T = \frac{1}{s} v^T A^{-1} \quad \boxed{\hat{v} = \frac{1}{s} (A^{-1})^T v} \quad (9)$$

Plug $\hat{v}^T$ into

$$\boxed{\begin{aligned} B &= A^{-1}\left(I_n - u\hat{v}^T\right) = \\ &= A^{-1}\left(I_n - u \frac{1}{s} v^T A^{-1}\right) \end{aligned}} \qquad (10)$$

Now $\tilde{A}^{-1}$ is completely specified.

Now we are back to matrix and vector norms.

(Fact): $\|A\|_2 = \sup_{\|x\|_2 = 1} \|Ax\|_2$

$\|A\|_2 = \max_i |\sigma_i|$, where σ_i are the singular values of A.

(these are e-values of AA^T)

Condition # and error analysis.

1. Perturb A^{-1} while solving $A\bar{x} = \bar{b}$

$A^{-1} \approx B$

Perturbed solution $\tilde{x} = B\bar{b}$

$$\|\bar{x} - \tilde{x}\| = \|\bar{x} - B\bar{b}\| = \|\bar{x} - BA\bar{x}\| =$$

$$= \|(I - BA)\bar{x}\| \leq \|I - BA\| \|\bar{x}\|$$

$\Downarrow$

relative perturbation:

$$\frac{\|\bar{x} - \tilde{x}\|}{\|\bar{x}\|} \leq \|I - BA\|$$

2. Perturb the r.h. side $\bar{b}$

$\bar{b} \simeq \tilde{b}$

Let $\tilde{x}$ solve $A\tilde{x} = \tilde{b}$

$$\|\bar{x} - \tilde{x}\| = \|A^{-1}\bar{b} - A^{-1}\tilde{b}\| =$$

$$= \|A^{-1}(\bar{b} - \tilde{b})\| \leq \|A^{-1}\| \|\bar{b} - \tilde{b}\|$$

We can also write

$$\|\bar{x} - \tilde{x}\| \leq \|A^{-1}\| \frac{\|A\bar{x}\|}{\|\bar{b}\|} \|\bar{b} - \tilde{b}\| =$$

$$= \|A^{-1}\| \|A\bar{x}\| \frac{\|\bar{b} - \tilde{b}\|}{\|\bar{b}\|} \leq$$

$$\leq \|A^{-1}\| \|A\| \|\bar{x}\| \frac{\|\bar{b} - \tilde{b}\|}{\|\bar{b}\|}$$

$$\Downarrow$$

$$\frac{\|\bar{x} - \tilde{x}\|}{\|\bar{x}\|} \leq \varkappa(A) \frac{\|\bar{b} - \tilde{b}\|}{\|\bar{b}\|}$$

where $\varkappa(A)$ is the <u>condition</u> <u>number</u> of A

$$\boxed{\varkappa(A) = \|A^{-1}\| \|A\|}$$

Note: $\varkappa(A) \geq 1$.

Indeed: $1 = \|I\| = \|A A^{-1}\| \leq$

$$\leq \|A\| \, \|A^{-1}\| = \varkappa(A)$$

Danger: large $\varkappa(A) \gg 1$.

$\boxed{\text{Ex}_1}$ $$A = \begin{bmatrix} 1 & 1+\varepsilon \\ 1-\varepsilon & 1 \end{bmatrix}, \quad \varepsilon > 0.$$

$$\det A = 1 - (1+\varepsilon)(1-\varepsilon) = 1 - (1-\varepsilon^2) = \varepsilon^2 \neq 0$$

$$A^{-1} = \varepsilon^{-2} \begin{bmatrix} 1 & -1-\varepsilon \\ -1+\varepsilon & 1 \end{bmatrix}$$

$$\|A\|_\infty = 2+\varepsilon, \quad \|A^{-1}\|_\infty = \varepsilon^{-2}(2+\varepsilon)$$

$$\varkappa(A) = \frac{(2+\varepsilon)^2}{\varepsilon^2} > \frac{4}{\varepsilon^2}$$

if $\varepsilon = 10^{-2}$, the $\varkappa(A) \geq 40.000$.